

Empirical Proof Record: VALIDATED

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Proof ID: e8e96b1fed7ffbb72c3c33492b24327c...

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1. Claim

Under sustained load, the substrate stays cognitively alive: (for every cycle in the trajectory, $\Phi \geq 0.05$). The reported floor is the lowest- Φ cycle across the whole run.

- **Operationalisation:** Stream 3 passes over 8 substantive stimuli through Kimera’s entity target (Takwin); read the per-cycle “phi”: `phi_floor_observed` = $\min \Phi$ over all cycles; the LTL invariant holds iff $\min \Phi \geq \varphi_floor$.
- **Threshold:** `phi_floor >= 0.05 phi`
- **H0:** H0: $\min \Phi < 0.05$ — Φ collapses below the non-collapse floor somewhere in the run
- **H1:** H1: $\min \Phi \geq 0.05$ — Φ stays above the collapse floor across the whole trajectory (substrate stays cognitively alive under sustained load)

2. Verdict

VALIDATED — observed 0.5997059933291004

Φ floor 0.5997 over 24 cycles (3 passes $\times$ 8 stimuli); mean Φ 0.6496, band [0.5997, 0.7148], stdev 0.0348; non-collapse rate 100.0%; 0 cycles produced no Φ ; worst cycle: stimulus 5 @ cycle 13 = 0.5997

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.flow.phi_stability	phi_floor	0.5997059933291004	—	ophamin 0.81.0

4. Pre-registration

- Registered at: 2026-05-21T23:30:26.204684+00:00
- Config hash: fc0d6fa2b433540d0fc987795a9290da...
- Data hash: f40e027c9d1b04da2ab75d1781c5c3f5...

5. Signature

1ec2928741ef5abd0fdd7baf791d57bd1c63414a53c7fed1a29b4f2d36010b71